

Churn Insights: Executive Summary



Churn analysis is the process of figuring out why customers stop doing business with you. This project analyses **21 customers** across three linked tables (customer, subscription, support). Following the SaaS definition, a customer has **churned** when the subscription is cancelled.

28.57%

Churn rate

6 of 21 customers

71.43%

Retention rate

15 of 21 customers

18.85

ARPU

avg monthly charge

73.94

Revenue lost

monthly charges churned

1,550

Avg tenure

days

19.05%

Escalation rate

4 of 21 customers

What the analysis shows – Who, Why, When:

- **"Who": Basic-plan** customers have the highest churn (60.00%), and customers acquired through **Referral** churn at 83.33% (Organic: 0%). By state, **Karnataka** (100%) and **Meghalaya** (66.67%) are highest.
- **"Why": Escalations** show a strong correlation with churn (0.77). Contract type also matters: monthly contracts are linked to higher churn (correlation -0.52 for annual vs monthly).
- **"When":** All cancellations fall in **2024**; churn peaks in **September 2024** (2 customers), with October and November each losing 1 more.
- **Revenue:** 73.94 of monthly charges belongs to churned customers. Premium plans bring the most revenue (218.93) and churn the least (14.29%).



Data & Preparation

Database **customer_churn** (SQLite) holds three tables, linked on **customerid**. Each was cleaned, then joined into one analysis table.

Table	Rows	Cleaning steps performed
db_customer	21	Renamed <i>name</i> → <i>customer_name</i> ; dropped <i>interests</i> and <i>pincode</i> ; <i>dob</i> → datetime; gender standardised (Men/Women → Male/Female); 3 missing <i>country</i> values filled using a state → country mapping
db_subscription	21	<i>subscription_start_date</i> , <i>renewal_date</i> , <i>cancellation_date</i> → datetime; new column churn_flag created from <i>cancellation_date</i>
db_support	9 → 7	Dropped <i>col_1</i> and <i>comment</i> ; <i>complaint_date</i> → datetime; complaints counted per customer, then reduced to one row per customer (7 unique customers)

Joining the tables and feature engineering

- Subscription, customer and support tables were left-joined on *customerid*. The first join (before fixing the support table) gave 23 rows; after the fix the table is **21 rows × 21 columns**.
- **tenure_days** was added (cancellation – start for churned customers; today – start for active ones).
- **churn_risk** was created from *churn_score*: high (> 70), mid (> 50 to 70), low (≤ 50), giving the final 23-column analysis table.

Churn Definition & Core KPIs



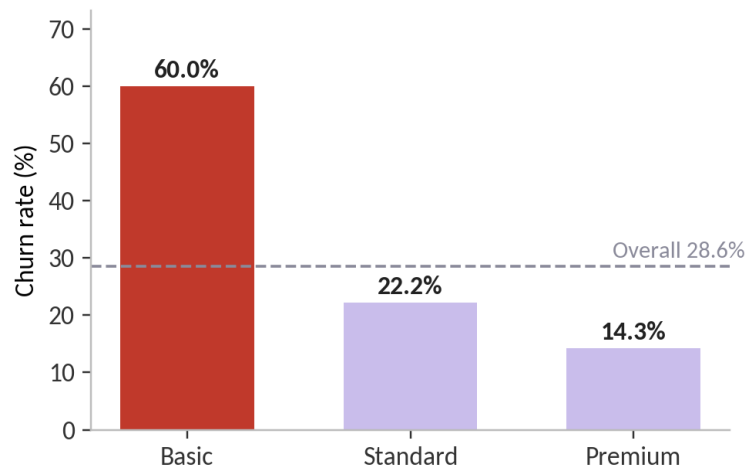
Churn definition (SaaS): subscription cancelled → **churn_flag = 1**. These are the KPIs computed in the analysis.

Metric	Value	How it was computed	Reading
Churn rate	28.57%	$\text{mean}(\text{churn_flag}) \times 100$	6 of 21 customers cancelled
Retention rate	71.43%	$100 - \text{churn rate}$	Customers who stayed
ARPU	18.85	$\text{mean}(\text{monthly_charges})$	Average revenue per user
Average tenure	1,550.14 days	Churned: cancellation – start; active: today – start	Average customer lifetime
Revenue lost to churn	73.94	$\text{sum}(\text{monthly_charges})$ where churn_flag = 1	Monthly charges of churned customers
Escalation rate	19.05%	share of customers with escalation = Y	4 of 21 customers
Escalation vs churn	0.77	$\text{correlation}(\text{escalations}, \text{churn_flag})$	Strong positive relationship

Who Is Leaving? Churn by Plan Type



Plan	Customers	Churned	Churn rate	Total monthly charges
Basic	5	3	60.00%	52.95
Standard	9	2	22.22%	123.91
Premium	7	1	14.29%	218.93

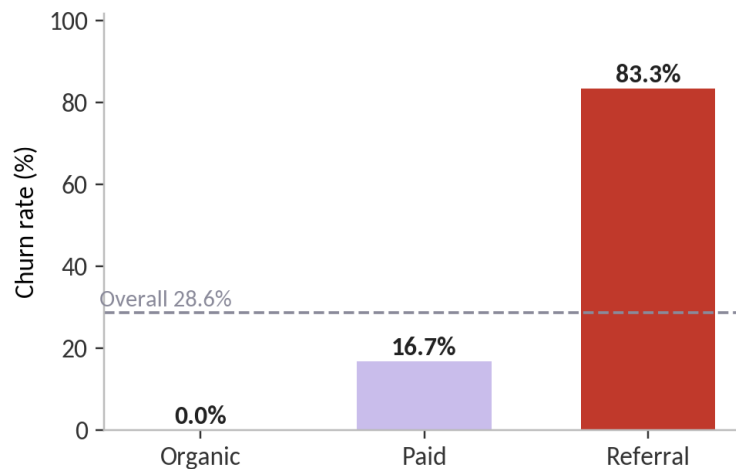


- **Basic has the highest churn (60.00%)** – 3 of its 5 customers cancelled.
- Churn falls as the plan tier rises: Basic 60.00% → Standard 22.22% → Premium 14.29%.
- The heatmap agrees: plan type has a negative correlation with churn (-0.36).
- **Premium** brings the highest revenue (218.93) with the lowest churn; Basic brings the lowest revenue (52.95).

Who Is Leaving? Churn by Channel



Subscription type	Customers	Churned	Churn rate	Total monthly charges
Organic	9	0	0.00%	145.91
Paid	6	1	16.67%	174.94
Referral	6	5	83.33%	74.94



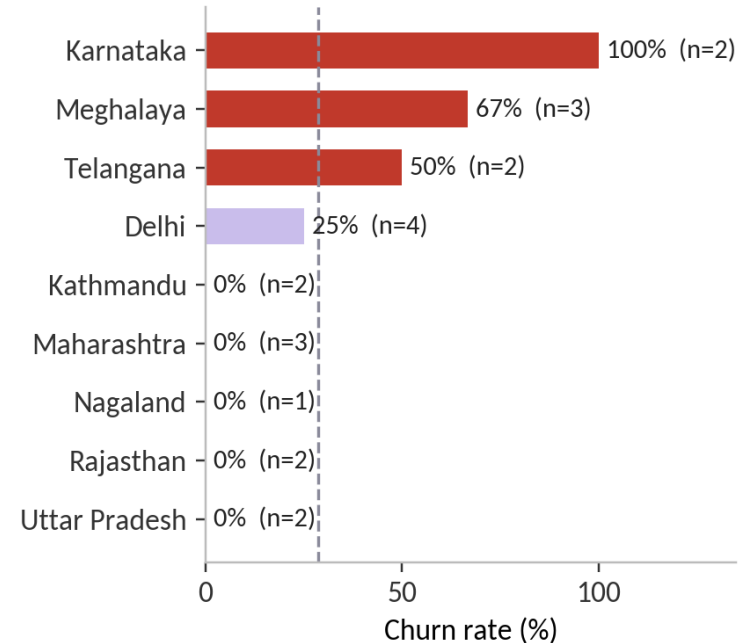
- **Referral has by far the highest churn (83.33%)** – 5 of its 6 customers cancelled.
- **Organic customers did not churn at all (0.00%).**
- **Paid customers churn at 16.67%** and bring the highest revenue (174.94).
- Referral customers bring the lowest revenue (74.94).

Where Is Churn? Churn by State



State	Customers	Churned	Churn rate	Revenue
Karnataka	2	2	100.0%	20.98
Meghalaya	3	2	66.7%	42.97
Telangana	2	1	50.0%	30.98
Delhi	4	1	25.0%	52.96
Uttar Pradesh	2	0	0.0%	115.98
Maharashtra	3	0	0.0%	50.97
Rajasthan	2	0	0.0%	36.98
Nagaland	1	0	0.0%	22.99
Kathmandu	2	0	0.0%	20.98

- **Karnataka** (100%) and **Meghalaya** (66.7%) have the highest churn; Telangana is at 50%.
- **Delhi** has the most customers (4) and a 25% churn rate.
- **Uttar Pradesh** has the highest revenue (115.98) and no churn.
- Five locations had no churn at all.
- Each state has only 1-4 customers (n in the chart).



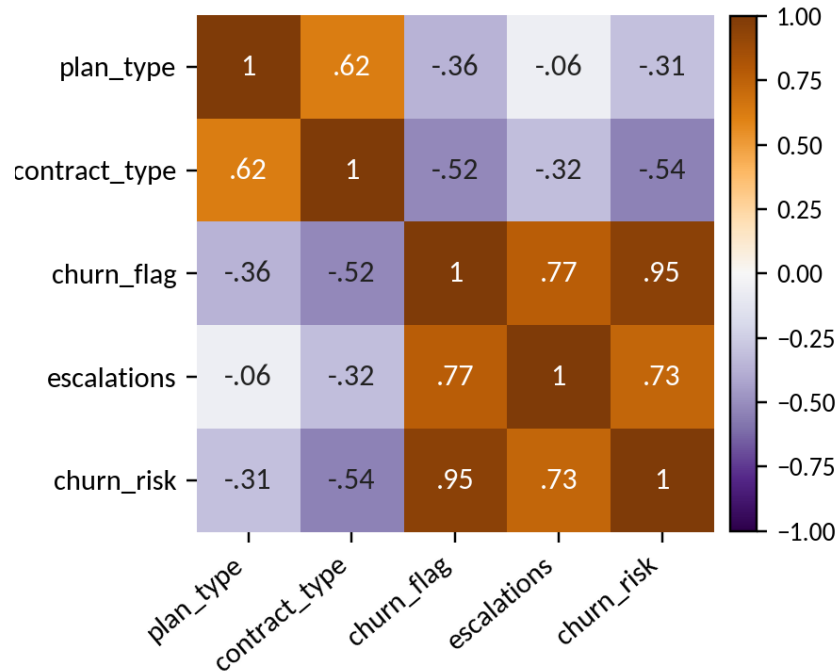
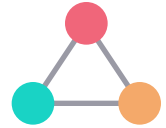
Why Do They Leave? Support & Escalations



- The support table has **9 complaints from 7 customers**; it was reduced to one row per customer before joining.
- **Escalation rate is 19.05%** – 4 of the 21 customers have an escalated complaint.
- Escalation has a **strong positive correlation with churn (0.77)** – customers with escalations are much more likely to have cancelled.
- Escalation also correlates positively with the churn_risk band (0.73).
- Escalation has almost no relationship with plan type (-0.062).

Escalation metric	Value
Escalation rate	19.05%
Correlation with churn_flag	0.77
Correlation with churn_risk	0.73
Correlation with contract_type	-0.32
Correlation with plan_type	-0.062

Why Do They Leave? Drivers & Correlations

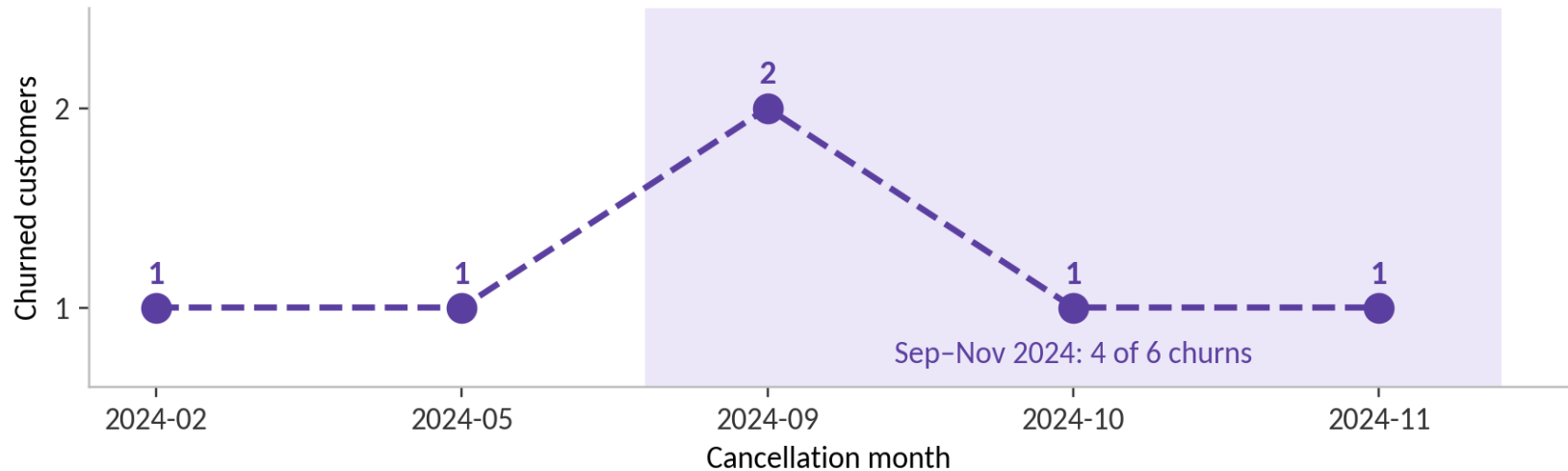


Correlation with churn_flag (ordered encoding)

- **churn_risk +0.95** – the churn_score bands track actual churn closely
- **escalations +0.77** – the strongest behavioural driver
- **contract_type -0.52** – annual contracts churn less than monthly
- **plan_type -0.36** – higher tiers churn less
- **plan_type ↔ contract_type +0.62**: higher plan tiers tend to go with annual contracts

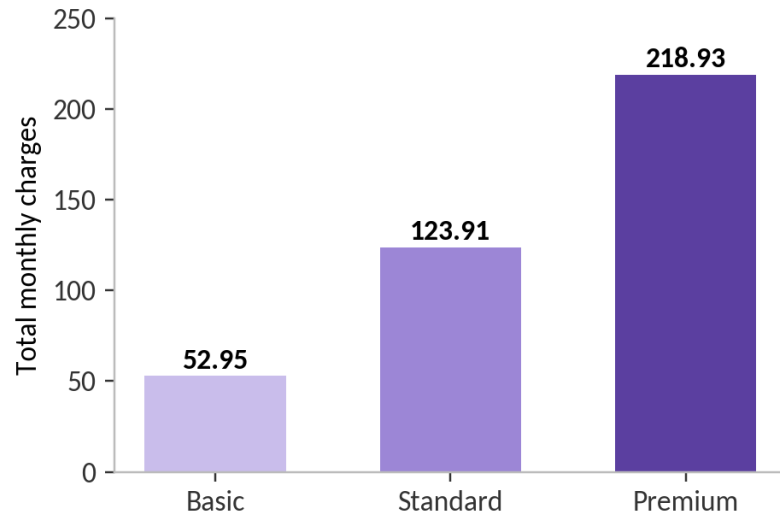
Encoding lesson from the project: the first heatmap used alphabetical category codes, which gave churn_risk vs churn_flag = -0.87 and contract_type vs churn_flag = +0.52. Re-encoding in logical order (Basic < Standard < Premium; Monthly < Annual; low < mid < high) corrected these to **+0.95** and **-0.52**.

When Do They Leave? Churn Trend



- All cancellations fall in **2024**, across 5 months: Feb (1), May (1), **Sep (2)**, Oct (1), Nov (1).
- **September 2024 is the peak** – the only month with 2 churned customers.
- Churn is concentrated in the second half of the year: Sep, Oct and Nov account for 4 of the 6 cancellations.
- No cancellations appear before February 2024 or after November 2024.

Revenue Impact



- **73.94** of monthly charges belongs to churned customers.
- **ARPU is 18.85** per customer.
- By plan, revenue rises with the tier: Basic 52.95, Standard 123.91, **Premium 218.93**.
- By channel, **Paid** brings the most revenue (174.94) and **Referral** the least (74.94).
- By state, **Uttar Pradesh** is the top revenue state (115.98) with no churn.
- The catplot shows one low-risk Premium customer with a much higher monthly charge than everyone else.

Summary: Premium customers generate the most revenue and churn the least, while Basic customers generate the least revenue and churn the most. Referral is the weakest acquisition channel on both measures.